중앙대학교

Mamba-based modulated fusion model for video moment retrieval

(2026-Scientific Reports)

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VMR (Video Moment Retrieval)

VMR (Video Moment Retrieval)

기존 VMR 모델의 흐름

1. **propose-and-rank paradigm : two-stage**
   1-1. Propose 단계: 전체 비디오 내에서 정답이 될 가능성이 있는 여러 개의 시간적 구간 후보
   1-2. Rank 단계: cross-modal matching model을 사용해 앞서 생성된 후보 구간들이 텍스트 쿼리와 얼마나 잘 일치하는지 평가 + 가장 적합한 순서대로 순위 매김
   한계: 초기 제안에 대한 지나친 의존, 경계 예측의 부정확성, 비효율성

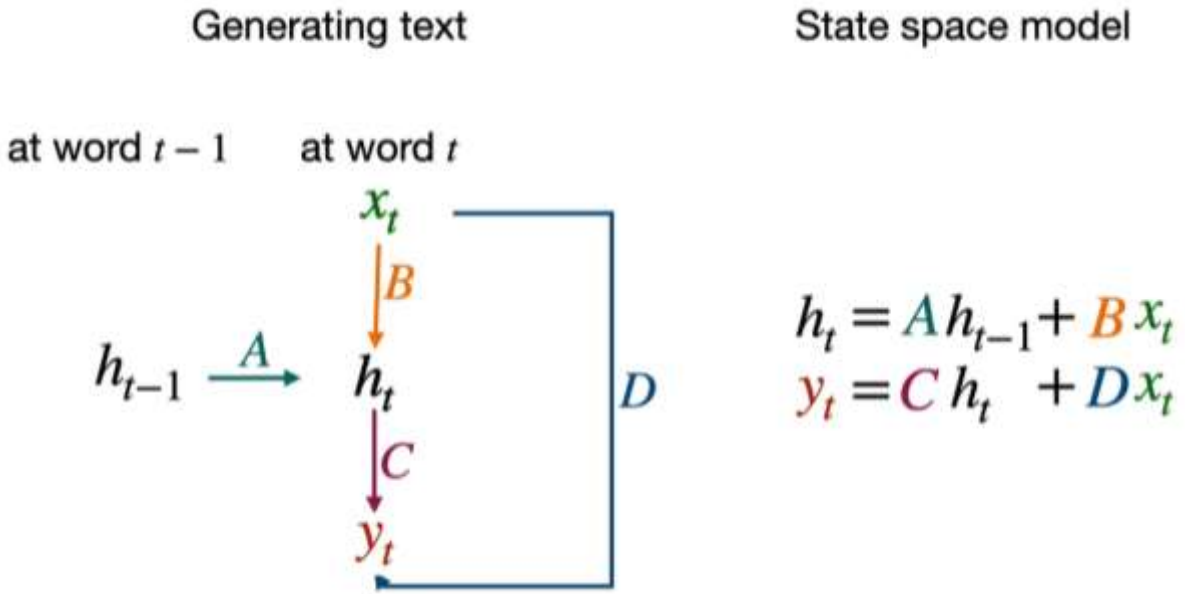
2. **proposal-free : end-to-end**
   Transformer 기반으로 정확한 시작점과 끝점(시간적 경계)을 중간 단계 없이 직접 예측
   한계: 긴 시간적 의존성을 모델링하기 어려움

시간 흐름을 더 잘 다루는 구조가 필요함

SSM (State Space Models)

SSM
특히 순서가 있는 데이터를 다룰 때 Transformer를 대체할 대안으로 떠오른 기술
장점: 긴 sequence를 효율적으로 모델링

$$h'(t) = Ah(t) + Bx(t)$$
$$y(t) = Ch(t)$$

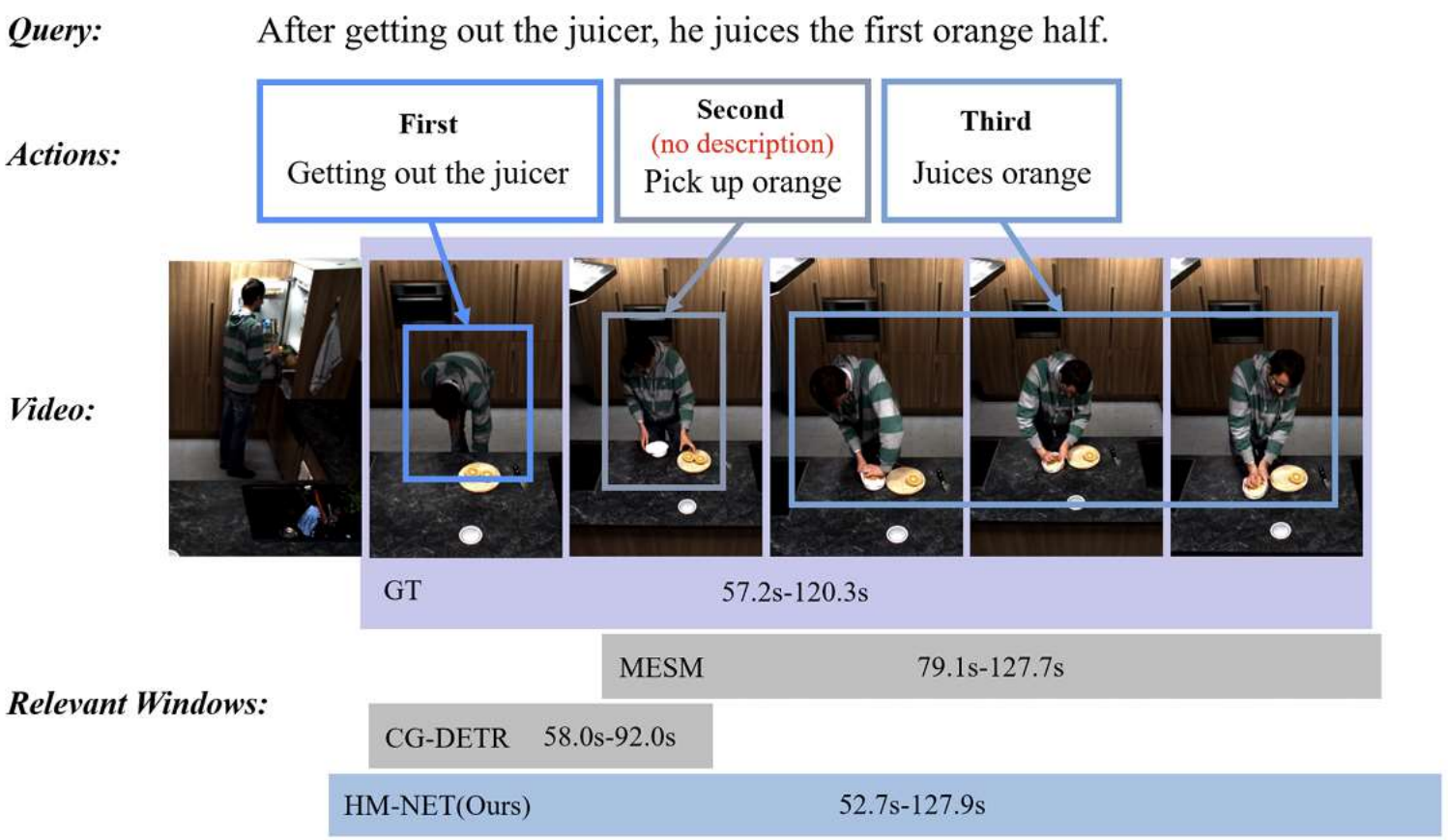


Mamba
selective mechanism 추가
  중요한 입력 → state에 강하게 반영
  덜 중요한 입력 → 약하게 반영하거나 무시
  이미 오래된 불필요한 정보 → 잊음
  필요한 과거 정보 → 유지
linear complexity

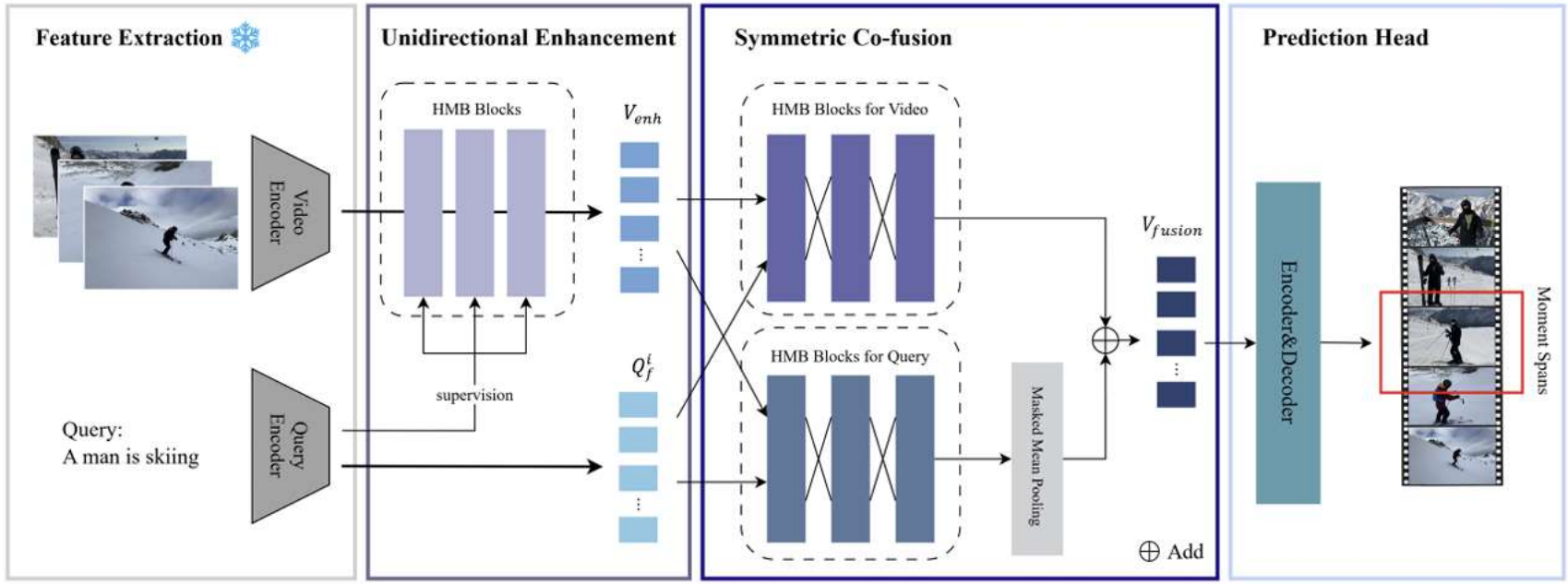
HM-Net

Hybrid Mamba Network

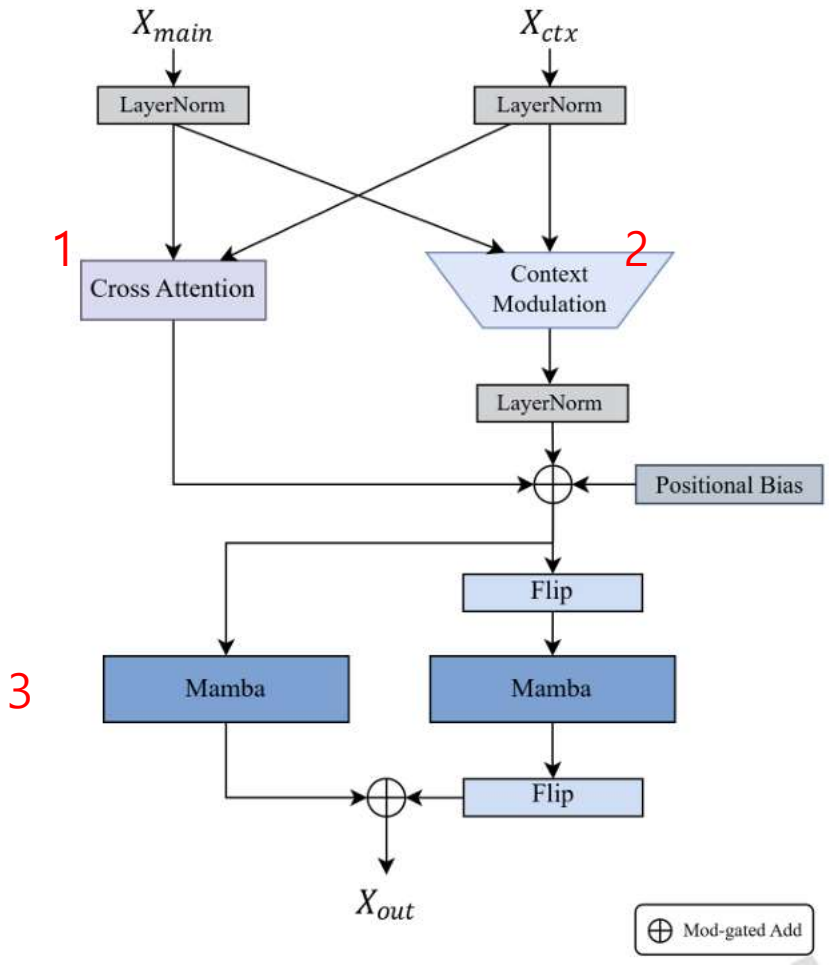
HM-Net: Hybrid Mamba Network



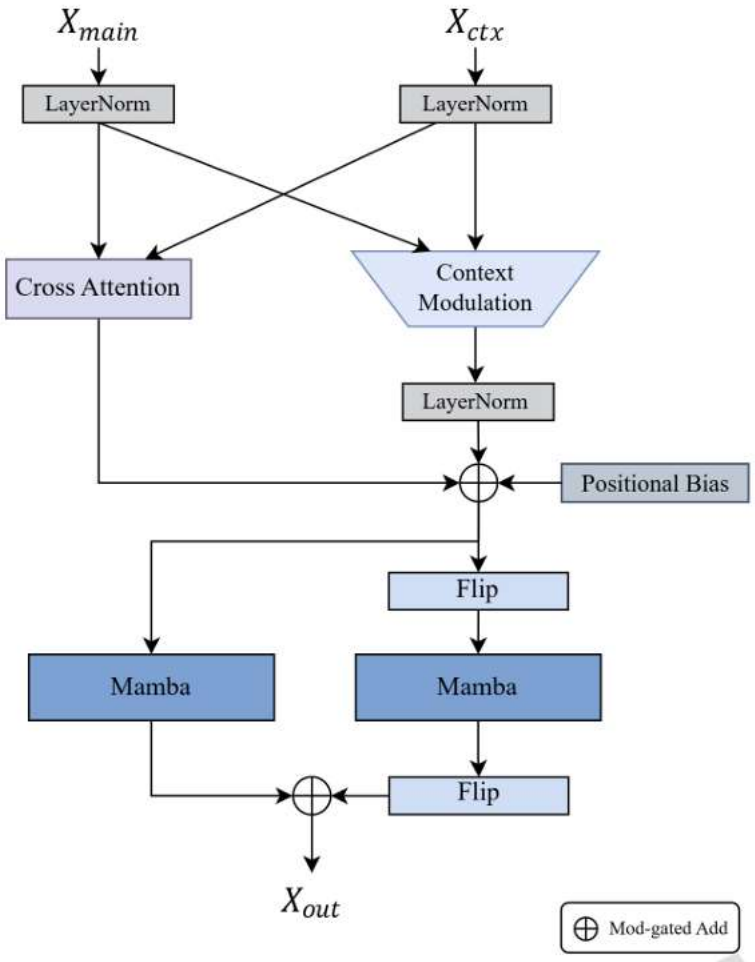
HM-Net


Feature Extraction
Video Encoder
Query:
A man is skiing
Query Encoder
Unidirectional Enhancement
HMB Blocks
V_{enh}
supervision
Q_f^i
Symmetric Co-fusion
HMB Blocks for Video
HMB Blocks for Query
Masked Mean Pooling
V_{fusion}
⊕ Add
Prediction Head
Encoder&Decoder
Moment Spans

HMB (Hybrid Modulated Bi-Mamba Block)



HMB (Hybrid Modulated Bi-Mamba Block): Cross-Modal Alignment.



1

X_{main} : 업데이트 대상 $X_{main} \in R^{L_m \times d}$ (e.g., video features)
X_{ctx} : 참고 대상 $X_{ctx} \in R^{L_c \times d}$ (e.g., text features)

$$X_{attn} = CrossAttn(LN(X_{main}, LN(X_{ctx}), LN(X_{ctx})))$$

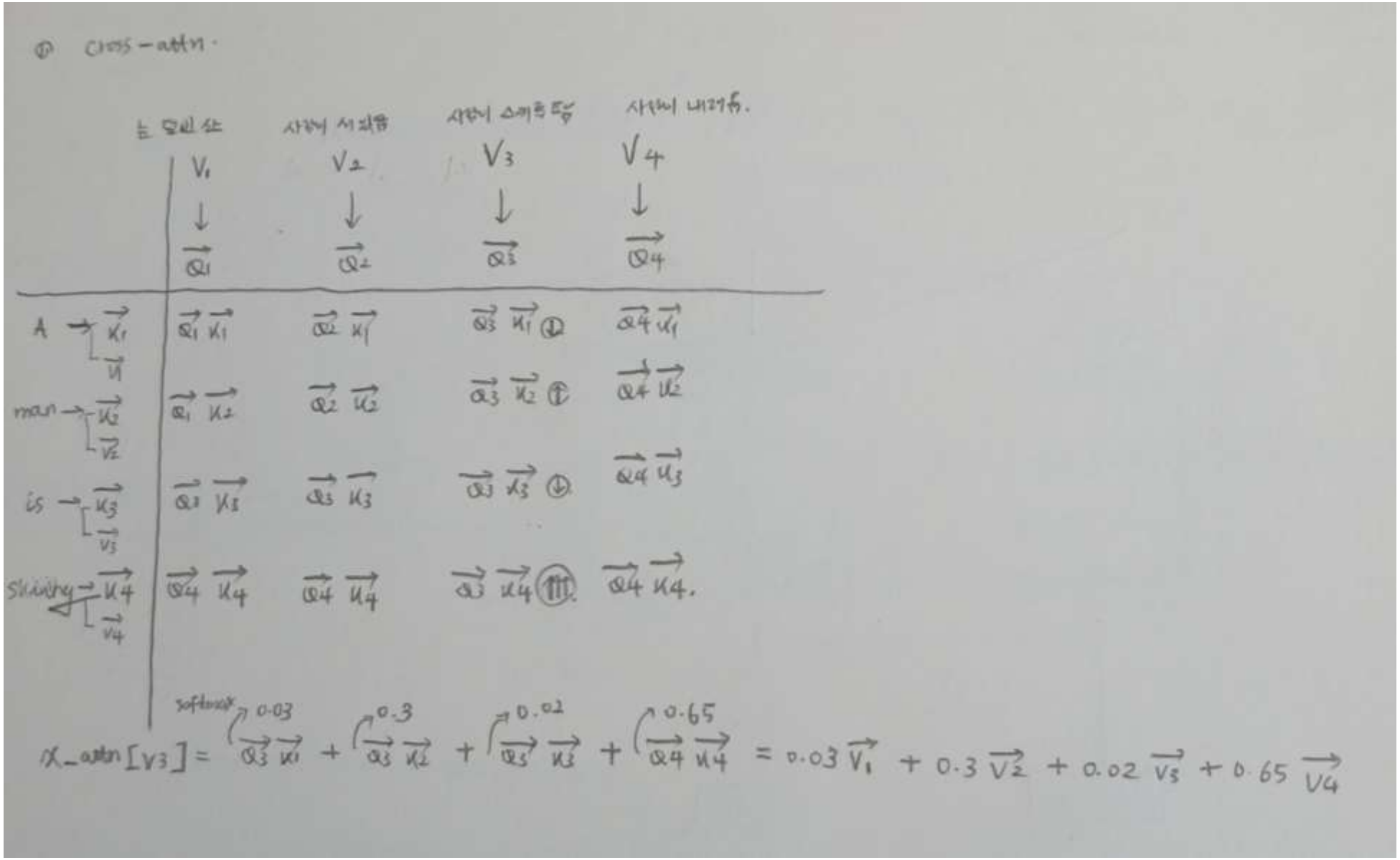
$$X_{attn} = CrossAttn(LN(X_{main}, LN(X_{ctx}), LN(X_{ctx})))$$

$$X_{attn} = CrossAttn\big(LN(X_{main}), LN(X_{ctx}), LN(X_{ctx})\big)$$



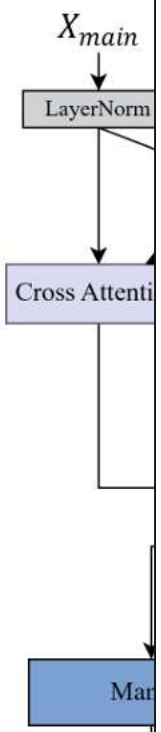
main sequence acts as the query
context sequence provides the keys and values
Q = X_main
K = X_ctx
V = X_ctx

LN(·): Layer Normalization


① cross-attn.
는 모델 쓰
사람이 시작을
사람이 스케치 중
사람이 내리침.
V1
V2
V3
V4
Q1
Q2
Q3
Q4
A
man
is
skiing
K1
K2
K3
K4
V1
V2
V3
V4
Q1 K1
Q2 K1
Q3 K1 ①
Q4 K1
Q1 K2
Q2 K2
Q3 K2 ①
Q4 K2
Q1 K3
Q2 K3
Q3 K3 ①
Q4 K3
Q4 K4
Q4 K4
Q3 K4 ①
Q4 K4.
softmax
0.03
0.3
0.02
0.65
X_attn[V3] = Q3 K1 + Q3 K2 + Q3 K3 + Q4 K4 = 0.03 V1 + 0.3 V2 + 0.02 V3 + 0.65 V4

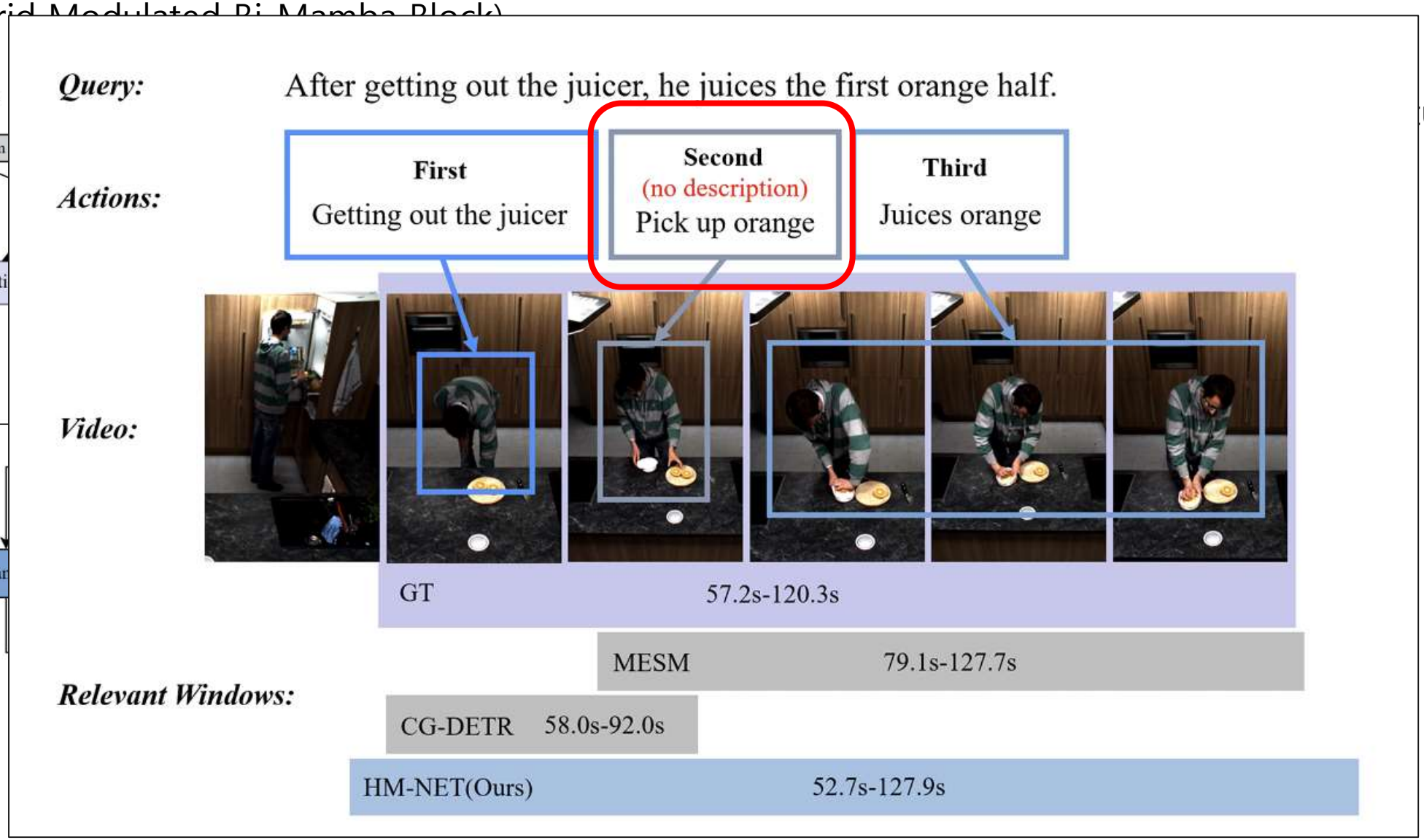
HM-Net

HMB (Hybrid Modulated Bi-Mamba Block)

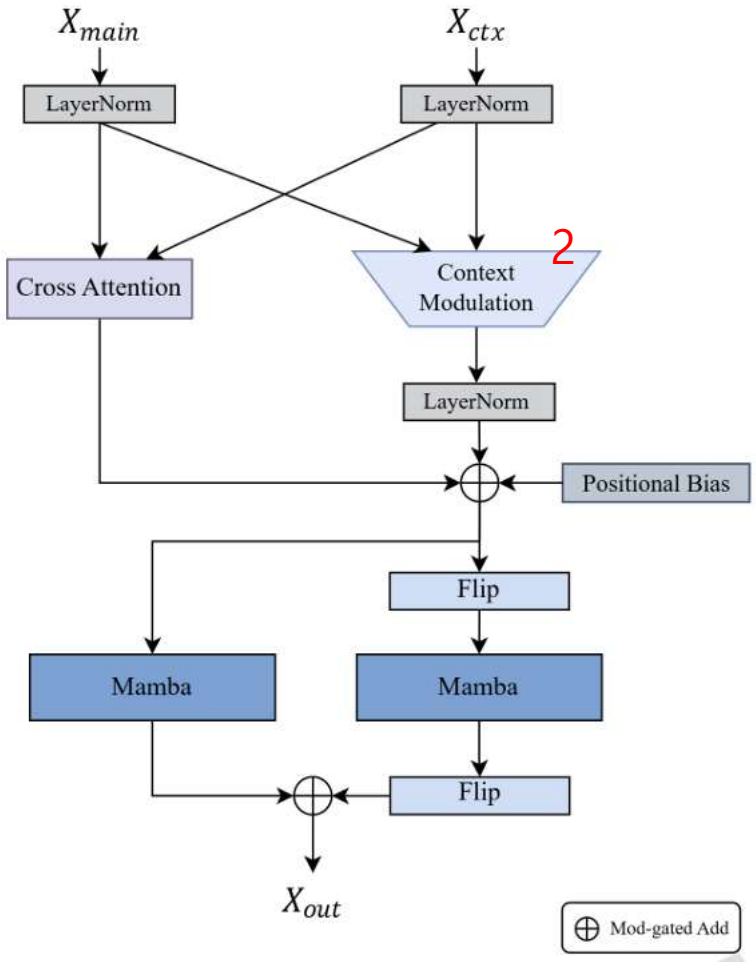


1

tures)



HMB (Hybrid Modulated Bi-Mamba Block): Hybrid Feature Modulation.



X_{main} : 업데이트 대상 $X_{main} \in R^{L_m \times d}$ (e.g., video features)
X_{ctx} : 참고 대상 $X_{ctx} \in R^{L_c \times d}$ (e.g., text features)

$$c_{\text{global}} = \frac{\sum_{i=1}^{L_c} M_{\text{ctx}}[i] \cdot X'_{\text{ctx}}[i]}{\sum_{i=1}^{L_c} M_{\text{ctx}}[i]}$$

$$S_{\text{mod}} = W_{\text{mod}}(c_{\text{global}})$$

$$X_{\text{fused}} = X'_{\text{main}} + X_{\text{attn}} + \tanh(\gamma_{\text{mod}}) \cdot S_{\text{mod}}$$

Where

$X'_{\text{main}} = LN(X_{\text{main}})$: normalized main-sequence input

X_{attn} : the output of the cross-modal interaction module

여기서 잠깐!

$$c_{\text{global}} = \frac{\sum_{i=1}^{L_c} M_{\text{ctx}}[i] \cdot X'_{\text{ctx}}[i]}{\sum_{i=1}^{L_c} M_{\text{ctx}}[i]}$$

M_{ctx} : padding mask

valid token: 실제 의미있는 token

문장 A: After getting out the juicer, he juices the first orange half.
문장 B: A man runs.
문장 C: The person cuts an onion on the table.



문장 A: [After, getting, out, the, juicer, he, juices, the, first, orange, half]
문장 B: [A, man, runs, PAD, PAD, PAD, PAD, PAD, PAD, PAD, PAD]
문장 C: [The, person, cuts, an, onion, on, the, table, PAD, PAD, PAD]

Mask: 각 token이 valid인지 invalid인지 표시
Ex) 문장 B
문장 B: [A, man, runs, PAD, PAD, PAD, PAD, PAD, PAD, PAD, PAD]
문장 B: [1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0]

mask가 1인 token만 더하고, mask가 1인 token 개수로 나눈다.

여기서 잠깐!

$$c_{\text{global}} = \frac{\sum_{i=1}^{L_c} M_{\text{ctx}}[i] \cdot X'_{\text{ctx}}[i]}{\sum_{i=1}^{L_c} M_{\text{ctx}}[i]}$$

M_{ctx} : padding mask

적용: [A, man, runs, PAD, PAD]

If) 각 token vector가 다음과 같다고 하자.
A = 2
man = 4
runs = 6
PAD = 0
PAD = 0

[1, 1, 1, 0, 0]

그냥 평균: (2 + 4 + 6 + 0 + 0) / 5 = 2.4
mask-weighted average: ((1×2 + 1×4 + 1×6 + 0×0 + 0×0) / (1+1+1+0+0)
= 12/3 = 4

여기서 잠깐!

$$c_{\text{global}} = \frac{\sum_{i=1}^{L_c} M_{\text{ctx}}[i] \cdot X'_{\text{ctx}}[i]}{\sum_{i=1}^{L_c} M_{\text{ctx}}[i]}$$

$$S_{\text{mod}} = W_{\text{mod}}(c_{\text{global}})$$
$$X_{\text{fused}} = X'_{\text{main}} + X_{\text{attn}} + \tanh(\gamma_{\text{mod}}) \cdot S_{\text{mod}}$$

Where

$X'_{\text{main}} = LN(X_{\text{main}})$: normalized main-sequence input

X_{attn} : the output of the cross-modal interaction module

W_{mod}

: c_{global}을 video feature에 더할 수 있는 형태로 바꿔주는 learnable matrix

예시

Xmain = video

Xctx = text

Xfused = video 원본 정보

+ video가 text를 cross-attention해서 얻은 정보
+ text 전체 의미를 video 쪽에 주입한 context modulation 정보

→ text 정보를 반영한 video feature sequence

eg. $d=3$

$C_{global} = [2, 1, 3]$ 1·3

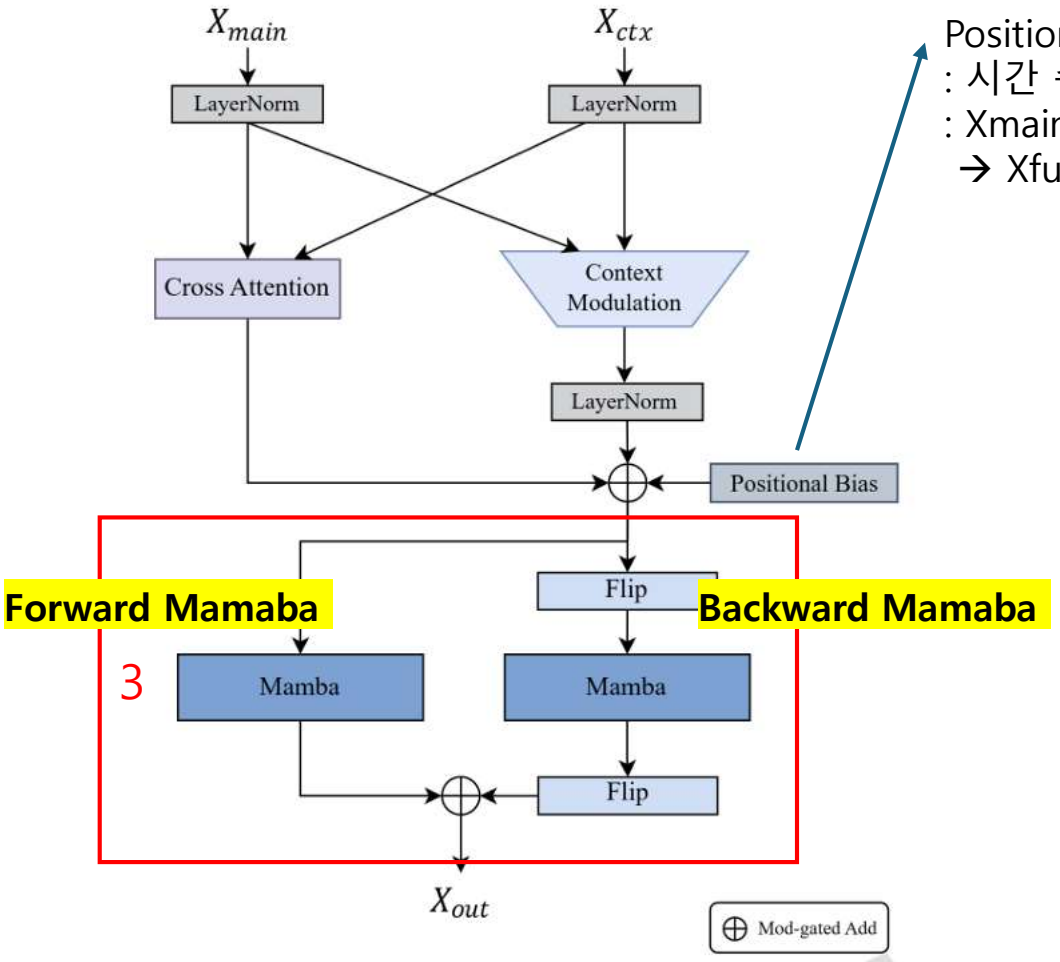
$$W_{mod} = \begin{pmatrix} [0.1 & 0.4 & 0.2] \\ [0.5 & 0.6 & 0.7] \\ [0.8 & 0.9 & 0.4] \end{pmatrix}$$

$$S_{mod} = (2, 1, 3) \begin{pmatrix} 0.1 & 0.4 & 0.2 \\ 0.5 & 0.6 & 0.7 \\ 0.8 & 0.9 & 0.4 \end{pmatrix} = (2\times0.1 + 1\times0.5 + 3\times0.8,\ 1\times0.4 + 1\times0.6 + 3\times0.9,\ 2\times0.2 + 1\times0.7 + 3\times0.4)$$

$$= (3.1, 4.1, 2.3)$$

$\therefore S_{mod} : [3.1, 4.1, 2.3]$

HMB (Hybrid Modulated Bi-Mamba Block): Bidirectional Sequence Modeling.

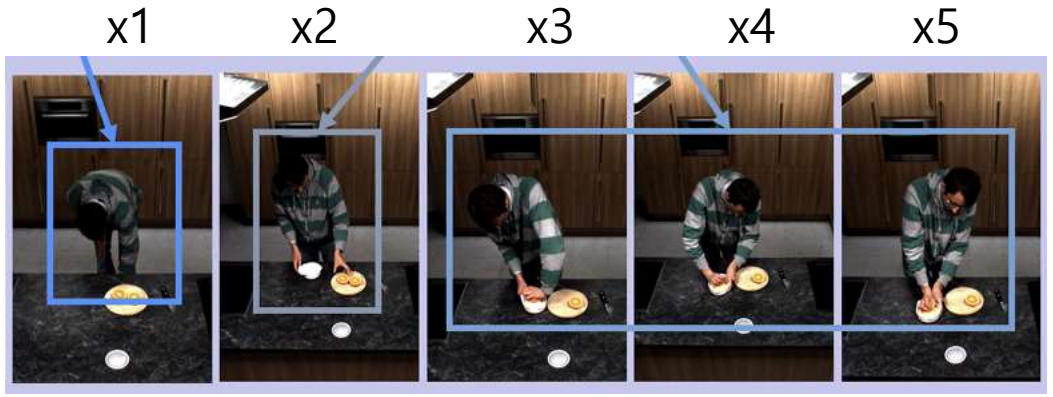


Positional Bias
: 시간 순서 정보(position information)를 보조적으로 넣어주는 장치
: Xmain의 길이(Lm)가 특정 threshold보다 작을 때만 활성화
→ Xfused에 위치 정보 추가

$$O_{\text{fwd}} = \text{Mamba}_{\text{fwd}}(\text{LN}(X_{\text{fused}}))$$ **Forward Mamaba**

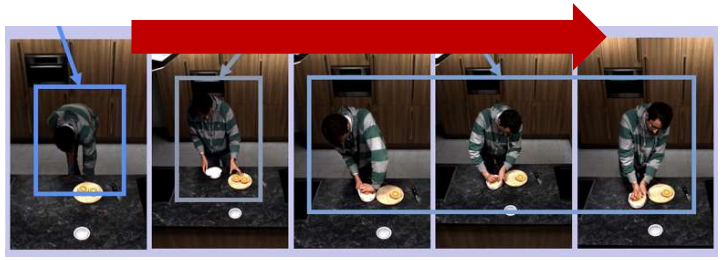
$$O_{\text{bwd}} = \text{Flip}(\text{Mamba}_{\text{bwd}}(\text{Flip}(\text{LN}(X_{\text{fused}}))))$$ **Backward Mamaba**

$$X_{\text{out}} = X_{\text{main}} + \text{DropPath}(X_{\text{fused}} + \tanh(\gamma_{\text{mamba}}) \cdot (O_{\text{fwd}} + O_{\text{bwd}}))$$



Forward Mamba:

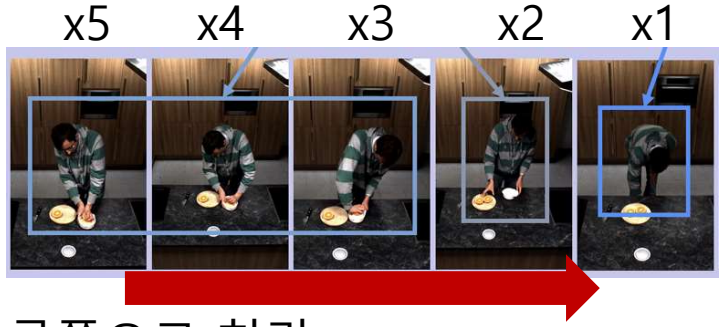
x1 — x2 — x3 — x4 — x5



Backward Mamba:

첫 Flip:

x5 — x4 — x3 — x2 — x1



Mamba는 여전히 왼쪽에서 오른쪽으로 처리:

→ → → → →

x5 — x4 — x3 — x2 — x1

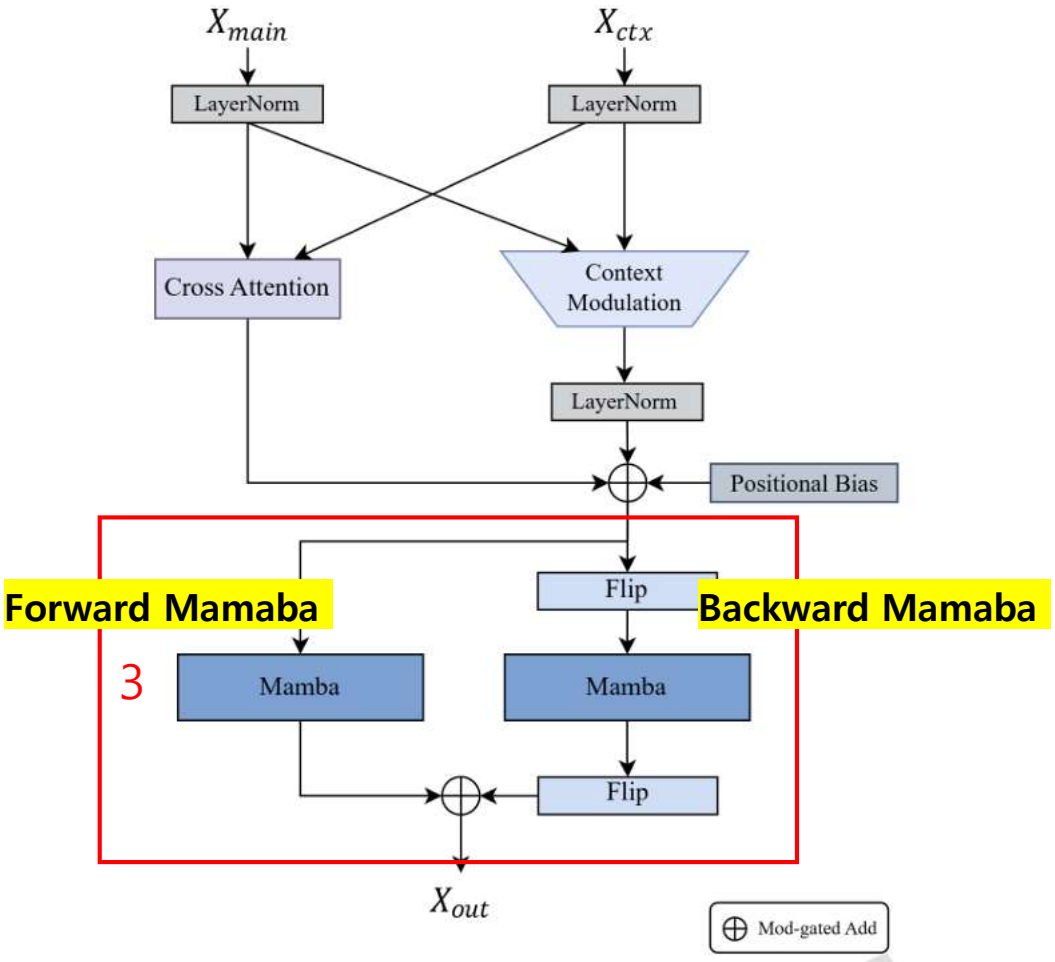
두 번째 Flip:

원래 시간 기준으로 보면:

← ← ← ← ←

x1 — x2 — x3 — x4 — x5

HMB (Hybrid Modulated Bi-Mamba Block)



$$O_{\text{fwd}} = \text{Mamba}_{\text{fwd}}(\text{LN}(X_{\text{fused}}))$$ **Forward Mamaba**

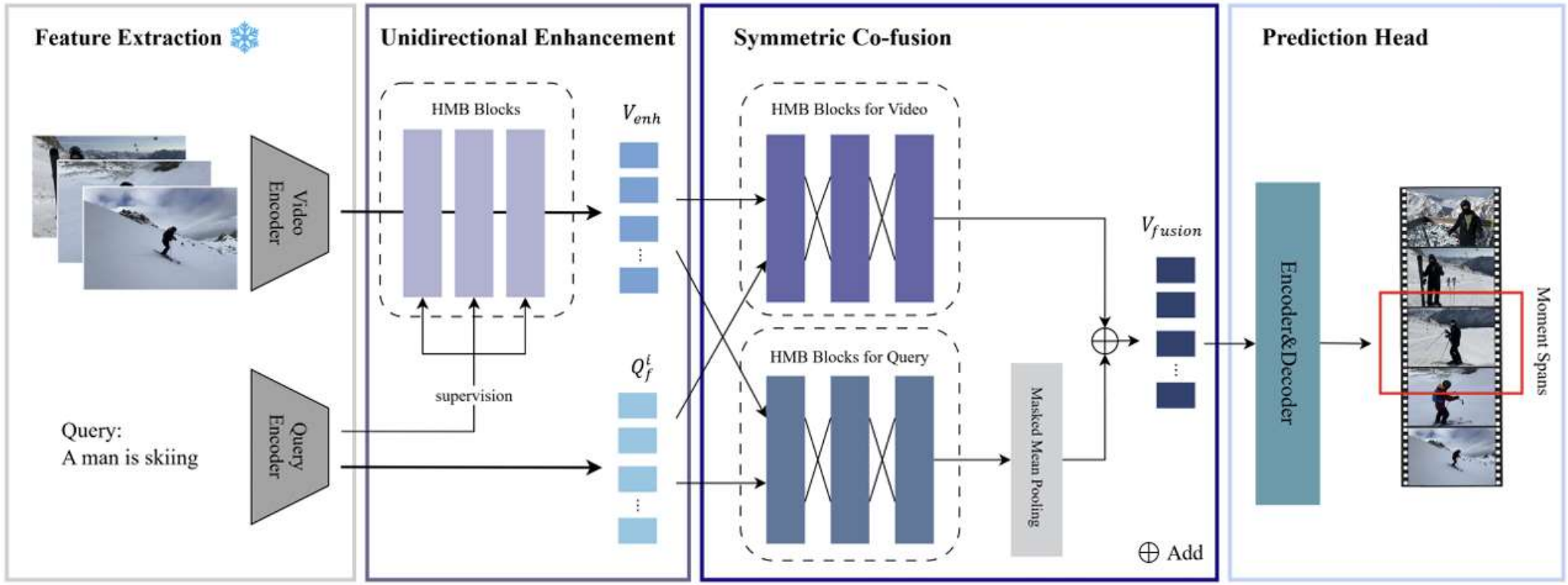
$$O_{\text{bwd}} = \text{Flip}(\text{Mamba}_{\text{bwd}}(\text{Flip}(\text{LN}(X_{\text{fused}}))))$$ **Backward Mamaba**

$$X_{\text{out}} = X_{\text{main}} + \text{DropPath}(X_{\text{fused}} + \tanh(\gamma_{\text{mamba}}) \cdot (O_{\text{fwd}} + O_{\text{bwd}}))$$

⊕ Mod-gated Add: learnable gate로 조절해서 더하기

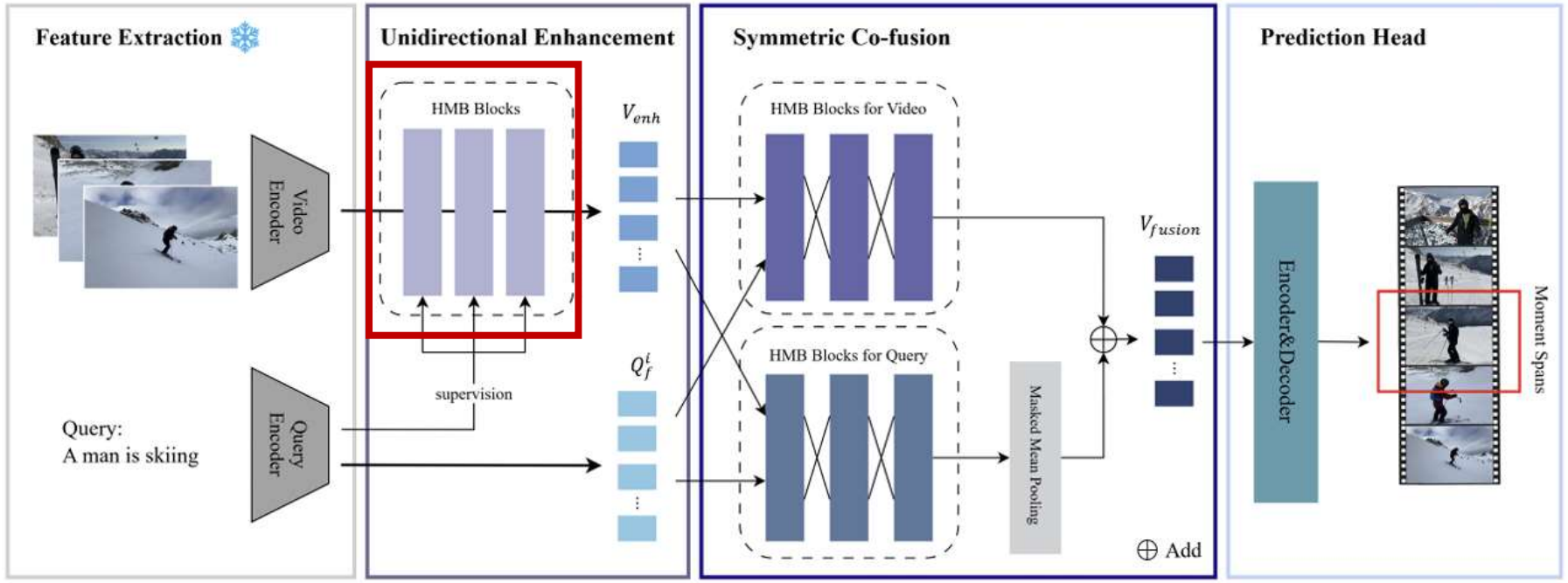
DropPath: $X_{fused} + tan\,h(\gamma_{mamba}) \cdot (O_{fwd} + O_{bwd})$
를 사용 할지 안할지

HM-Net


Feature Extraction
Video Encoder
Query:
A man is skiing
Query Encoder
Unidirectional Enhancement
HMB Blocks
V_{enh}
supervision
Q_f^i
Symmetric Co-fusion
HMB Blocks for Video
HMB Blocks for Query
Masked Mean Pooling
V_{fusion}
⊕ Add
Prediction Head
Encoder&Decoder
Moment Spans

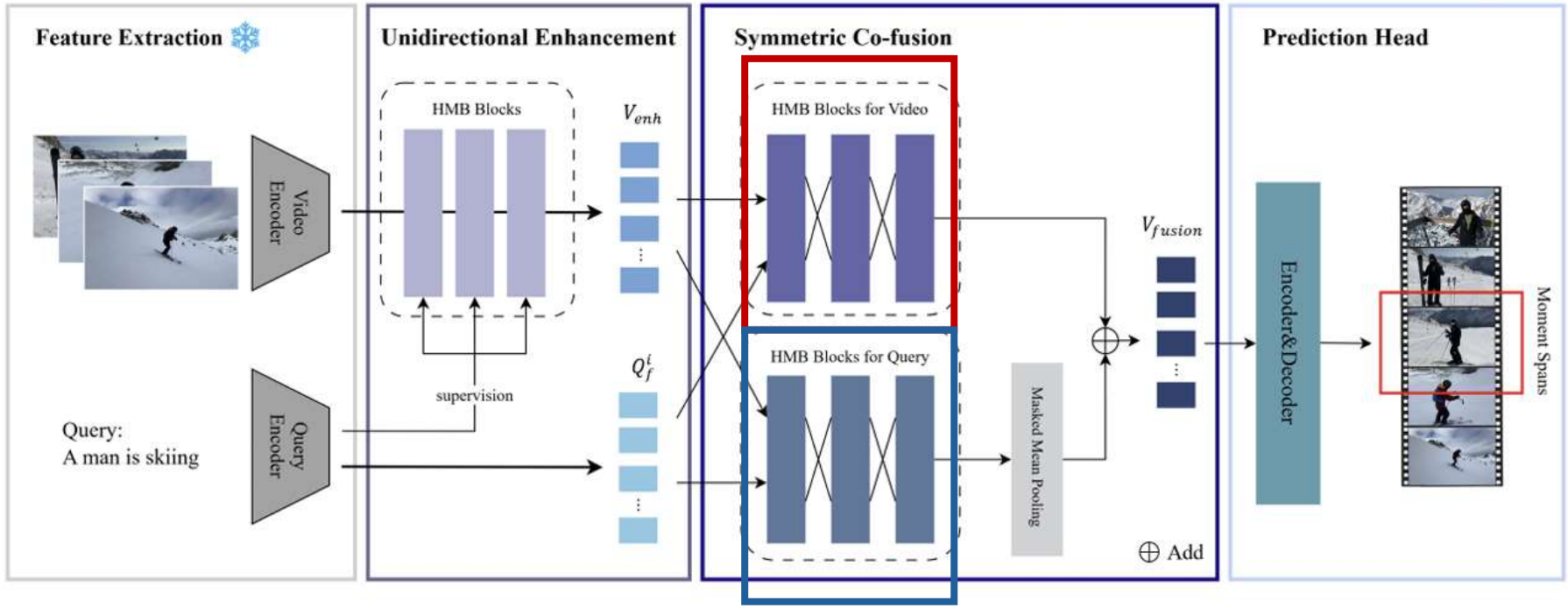
$$X_{main} = video$$
$$X_{ctx} = text$$

출력 : query-aware video feature



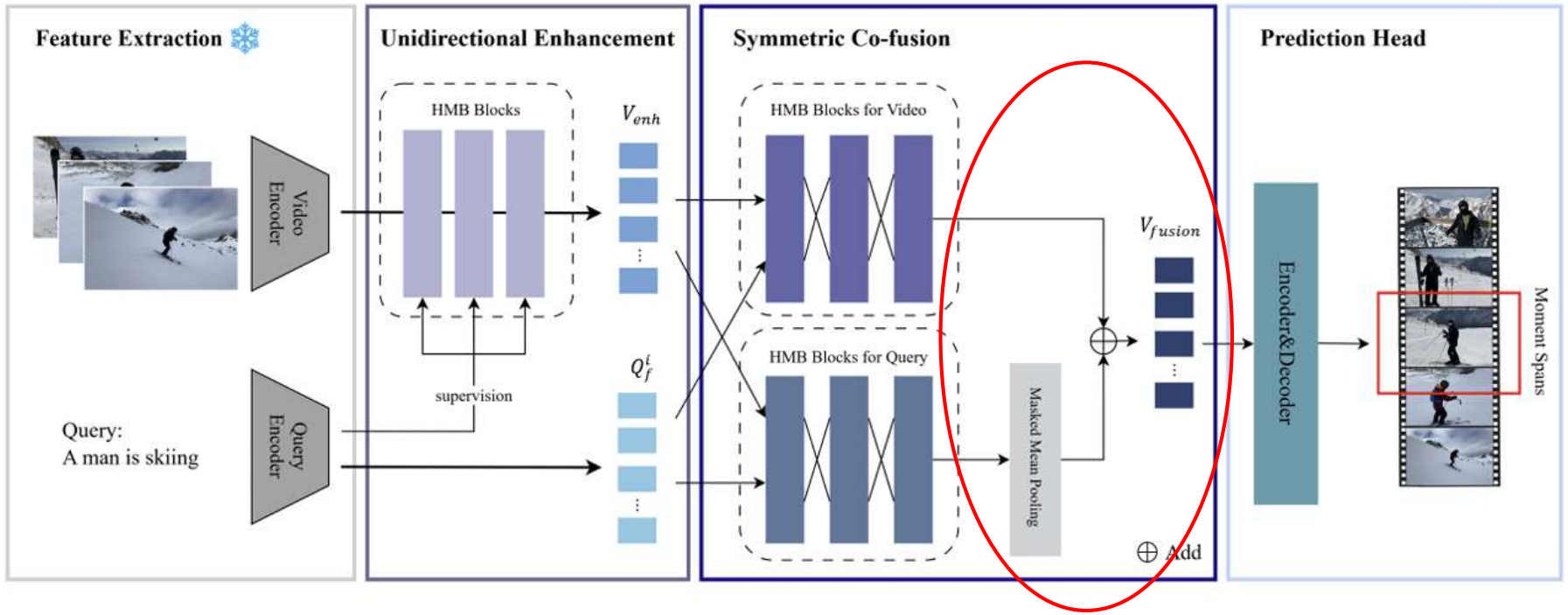
HM-Net

$$X_{main} = video \quad X_{ctx} = text \qquad V_f^{l+1} = HMB_v(V_f^{(l)}, Q_f^{(l)})$$



$$X_{main} = \text{text} \quad X_{ctx} = video \qquad Q_f^{l+1} = HMB_q(Q_f^{(l)}, V_f^{(l)})$$

HM-Net



updated video feature sequence
+
pooled global query vector
=
V_fusion

예시)
frame1 + "남자가 스키 타는 장면 기준으로 봐"
frame2 + "남자가 스키 타는 장면 기준으로 봐"
frame3 + "남자가 스키 타는 장면 기준으로 봐"

Experiment & Result

Experiment & Result

Experiment

Dataset	비디오/시각 feature	텍스트 feature
QVHighlights	SlowFast	CLIP
TACoS	C3D	GloVe

Metric:

R1@0.3
R1@0.5
R1@0.7
mAP@0.5
mAP@0.75
mAP Avg
mIoU

Table 1. Comparison results on the TACoS dataset. * denotes methods that rely on large-scale pretraining. For fairness, all methods are trained only on the target dataset. † denotes methods that additionally incorporate audio features.

Method	R1			mAP			mIoU
	@0.3	@0.5	@0.7	@0.5	@0.75	Avg.	
MMN[50]	37.19	26.39	13.02	-	-	-	-
Uni-VTG*[51]	48.62	35.39	17.76	40.17	17.57	20.26	32.69
LLMEPET*[28]	46.80	32.30	17.09	38.07	15.13	17.97	31.58
CICR[52]	53.09	38.94	20.09	42.15	15.79	19.68	35.76
MESM[39]	53.91	38.69	22.64	40.40	19.69	21.03	37.46
CG-DETR[38]	53.11	40.39	24.99	**45.16**	20.69	23.56	37.14
MRNet†[53]	55.14	39.32	22.24	-	-	-	38.77
HM-Net(Ours)	**55.71**	**43.16**	**27.94**	44.13	**23.15**	**25.01**	**40.04**

Table 2. Comparison results on the QVHighlights dataset. * denotes methods that rely on large-scale pretraining. For fairness, all methods are trained only on the target dataset.

Method	R1			mAP			mIoU
	@0.3	@0.5	@0.7	@0.5	@0.75	Avg.	
EaTR[54]	69.68	59.23	44.65	60.54	40.25	39.99	54.56
LLMEPET*[28]	63.61	47.10	27.48	50.37	26.67	27.98	44.49
Uni-VTG*[31]	72.58	59.81	42.71	51.15	33.80	33.39	54.24
QD-DETR[3]	74.52	62.06	46.32	62.17	41.34	40.96	57.38
MESM[39]	74.00	62.52	46.39	62.20	40.88	40.74	57.53
UVCOM[55]	74.00	**63.74**	47.68	62.17	43.23	41.52	58.01
CICR[52]	**75.35**	63.35	47.23	62.62	41.96	41.51	**58.26**
HM-Net(Ours)	74.77	62.77	**48.13**	**63.14**	**43.70**	**43.17**	57.88

Table 3. Ablation studies on the core components of HM-Net on the TACoS dataset. For Mamba, we compare the original bidirectional design (Bi) with a unidirectional variant (Uni) to validate the importance of bidirectional modeling.

Components			Performance Metrics		
Mamba	C-Attn	C-Pool	R1@0.5	mIoU	mAP@Avg.
	✓		38.69	37.46	21.03
✓ *Bi*			35.77	33.72	19.71
✓ *Bi*	✓		41.14	37.82	23.95
✓ *Uni*	✓	✓	41.06	38.52	23.12
✓ *Bi*	✓	✓	**43.16**	**40.04**	**25.01**

Table 4. Ablation studies on the depth of different modules on the TACoS dataset.

Module	Layers	R1@0.5	mIoU	mAP@Avg.
Enhancement	0	40.64	38.57	24.11
	1	41.31	38.39	**25.05**
	2	**43.16**	**40.04**	25.01
	3	42.54	39.43	24.15
Co-fusion	1	40.74	38.05	22.70
	2	**43.16**	**40.04**	**25.01**
	3	42.84	39.71	24.89

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Table 5. Quantitative comparison on long-video subsets. "Long": videos longer than 30 seconds for both TACoS and QVHighlights. "Super-long": videos longer than 150 seconds for TACoS.

Dataset	Duration	Method	mAP Avg.
TACoS	Long	MESM	15.81
		HM-Net (Ours)	**21.95**
	Super-long	Baseline	10.47
		HM-Net (Ours)	**12.75**
QVHighlights	Long	MESM	45.61
		HM-Net (Ours)	**50.61**

감사합니다